Discrete-Time Mixed Markov Latent Class Models

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This chapter deals with the situation where one discrete (or categorical) variable x, which may refer to attitudes, voting intentions, employment status or the occurrence of some sort of behaviour like buying, is measured at several (T) consecutive occasions, $x_1, x_2, \ldots x_T$, with realizations $i, j, \ldots, m$. In general T is rather small (say from three to six) and the sample size is rather large. In principle, x may be polytomous, but for ease of exposition we will consider dichotomous variables only. Such data, which may be compiled in contingency table form, may be available for one population only (see Table 8.1) or for several subpopulations (see Table 8.9). Analysis with Markov models presented in this chapter focuses on the types of change and stability that exist in these subpopulations.

These models are specifically tailored for the data at hand, taking into consideration the discreteness of both space and time. They may be formulated on the manifest level or extended to the latent level either by postulating several (unknown) types of change for a (sub)population or by incorporating measurement error. First developments in the field are mainly due to Wiggins (1955), a dissertation that later appeared in book form (Wiggins, 1973). Although conceptually appealing, Wiggins's work was plagued by several problems, most of which were tied to inefficient methods of parameter estimation. These problems were surmounted by Poulsen (1982) who showed how to perform maximum likelihood (ML) estimation of the parameters of the mixed Markov model (see below) and the latent Markov model (see below) by use of the expectation maximization (EM) algorithm of Dempster et al. (1977). In a parallel development Davies and Crouchley (1986) also described the mixed Markov model, restricting a continuous distribution to a set of latent classes.

Exogenous variables may be modelled by analyzing several subpopulations simultaneously (van de Pol and Langeheine, 1990). Spilerman (1972) may have been the first to analyze the effect of exogenous variables on a single Markov chain. He used a regression framework, which has been refined by Kelton and Smith (1991). Davies and Crouchley (1985), building on the econometric literature of discrete-choice models, included several exogenous variables and also modelled unobserved heterogeneity, but not with latent classes. Another difference with the approach that is adopted in the present chapter is that their model has no parameters for response uncertainty.

Our exposition will start with the most general model, the latent mixed Markov model for several subpopulations. From a didactic point of view it may not be optimal to start top-down with this model. However, we considerably simplify the notation used in the subsequent sections in doing so. Subsequently, we proceed bottom-up by starting with the simplest Markov model and ending with the most general model in the penultimate section. This enables us to focus on the strengths and weaknesses of the various models considered. Readers who feel the most general model to be too complicated should just try to get a cursory understanding and then continue with the simple Markov model in the next section.

The general latent mixed Markov model for several populations

Assume that repeated measurements of some discrete variable x are available for several (T) points in time for several (H) subpopulations. An example is given in Table 8.9. There we have H=2 subpopulations (males/females), x has J=2 categories (disabled/not disabled), and T=3 (1971,1972,1974). The data are thus given by an $H \times J^T$ table. The general latent mixed Markov model for several subpopulations (LMMS, where S stands for several subpopulations that are defined by additional external variables, like gender in Table 8.9) is in the case of three occasions given by

$$P_{hijk} = \gamma_h \sum_{s=1}^{S} \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \pi_{s|h}\, \delta^{1}_{a|sh}\, \rho^{1}_{i|ash}\, \tau^{21}_{b|ash}\, \rho^{2}_{j|bsh}\, \tau^{32}_{c|bsh}\, \rho^{3}_{k|csh} \tag{8.1}$$

with P_{hijk} the proportion in the population in cell (h,i,j,k). The model assumes that each subject belongs to one of one or more subpopulations. Membership of subpopulation h (h=1, ... , H) remains unchanged for all occasions. The proportion that belongs to subpopulation h is denoted by γ_h. All other parameters that will be described below are considered conditional on subpopulation h.

Each member of subpopulation h can belong to one of one or more (manifest or latent) Markov chains, i.e. parts of the (sub)population that have the same dynamics. A proportion $\pi_{s|h}$ in subpopulation h belongs to chain s (s = 1, ... , S). Hence the proportion in subpopulation h and chain s is $\gamma_h\, \pi_{s|h}$.

A member of subpopulation h and chain s is assumed to belong to one of A latent classes. The proportion in class a (a = 1, ... , A) at occasion 1, for subpopulation h and chain s, is denoted by $\delta^1_{a|sh}$. Hence the proportion in subpopulation h, chain s, and class a at occasion 1 is $\gamma_h\, \pi_{s|h}\, \delta^1_{a|sh}$.

The probability $\rho^1_{i|ash}$ of response i on occasion 1, given h, s and a, is assumed the same for all subjects in subpopulation h, chain s and class a. Hence, the proportion in subpopulation h, chain s, class a and category i at occasion 1 is $\gamma_h\, \pi_{s|h}\, \delta^1_{a|sh}\, \rho^1_{i|ash}$. If the model is manifest, the latent classes (indexed by a for occasion 1, b for occasion 2, etc.) and the manifest categories (i for occasion 1, j for occasion 2, etc.) are identical. Therefore, the response probabilities $\rho^1_{i|ash}$ are superfluous in a manifest model.

If the model applies to more than one occasion then, for subpopulation h, each member of chain s is assumed to behave according to the same transition probabilities, $\tau^{21}_{b|ash}$, from latent class a at time 1 to latent class b at time 2. As on occasion 1, the probability $\rho^2_{j|bsh}$ of response j on occasion 2, given h, s and b, is assumed to be the same for all subjects in subpopulation h, chain s and class b. Hence, the proportion in subpopulation h, chain s, class a and category i at occasion 1, and class b and category j at occasion 2, is $\gamma_h\ \pi_{s|h}\ \delta^1_{a|sh}\ \rho^1_{i|ash}\ \tau^{21}_{b|ash}\ \rho^2_{j|bsh}$.

Extension to more occasions is straightforward. Summing over chains and latent classes we obtain the proportion of subjects P_{hijk} in subpopulation h, category i at occasion 1, category j at occasion 2, and category k at occasion 3.

A sample gives an estimate of the P_{hijk} from which the γ, π, δ, ρ, and τ parameters may be estimated. Special cases of the general LMMS model considered in the subsequent sections have been estimated using the TURBO PASCAL™ program PANMARK (van de Pol et al., 1989). It computes maximum likelihood (ML) parameter estimates using a version of the EM algorithm and standard errors of the parameters by numerical differentiation of the likelihood (for details, see also van de Pol and de Leeuw, 1986; van de Pol and Langeheine, 1989; van de Pol and Langeheine, 1990).

For completeness, we note that in order to be uniquely defined all sets of parameters have to sum to unity, given any combination of variables that is conditioned on (e.g. $\Sigma_h\ \gamma_h = \Sigma_s\ \pi_{s|h} = \Sigma_a\ \delta^1_{a|sh} = \Sigma_i\ \rho^1_{i|ash} = \Sigma_b\ \tau^{21}_{b|ash} = 1$).

Models for one population

In order to demonstrate the advantages and disadvantages of several models we will first use a data set from a single population. The data, which are taken from Aaker (1970), are reproduced in Table 8.1. These data provide a record of family brand purchases over time. Each purchase is coded 2 if brand B is purchased, and 1 if another brand in the product class is purchased. Data are available for $T = 5$ occasions (or waves) for a sample of 988 families.

The simple Markov model

In order to pinpoint stability and change, data such as in Table 8.1 are often first analyzed using the simple Markov (M) model:

$$P_{ijk} = \delta^1_i\ \tau^{21}_{j|i}\ \tau^{32}_{k|j} \tag{8.2}$$

(As in the case of the general LMMS model we will give formulae by including $T = 3$ occasions only throughout. Extension to more occasions is straightforward.)

Table 8.1 *Contingency table[1] of brand purchase data*

Time $t=1$	$t=2$	$t=3$	$t=4$	$t=5$	Observed frequencies
1	1	1	1	1	464
1	1	1	1	2	31
1	1	1	2	1	26
1	1	1	2	2	12
1	1	2	1	1	28
1	1	2	1	2	9
1	1	2	2	1	6
1	1	2	2	2	5
1	2	1	1	1	49
1	2	1	1	2	5
1	2	1	2	1	7
1	2	1	2	2	2
1	2	2	1	1	12
1	2	2	1	2	5
1	2	2	2	1	12
1	2	2	2	2	10
2	1	1	1	1	79
2	1	1	1	2	11
2	1	1	2	1	10
2	1	1	2	2	8
2	1	2	1	1	12
2	1	2	1	2	8
2	1	2	2	1	3
2	1	2	2	2	12
2	2	1	1	1	31
2	2	1	1	2	5
2	2	1	2	1	7
2	2	1	2	2	9
2	2	2	1	1	25
2	2	2	1	2	12
2	2	2	2	1	15
2	2	2	2	2	58

[1]Categories: 2, purchase of brand B; 1, purchase of some other brand.
Source: Aaker, 1970

Model M (8.2) is derived from model (8.1) by taking into consideration the following issues. First, since we have one population only, there is no index h. Second, the simple Markov model assumes one chain only. There is therefore no index s. Thirdly, this chain is defined to be a manifest chain. Response probabilities, the ρs, are therefore superfluous, and the latent classes (a, b and c) coincide with the manifest categories (i, j and k). Hence the proportion of subjects with response sequence i, j and k is given by the probability of response i on occasion 1 (δ^1_i), multiplied by the probability of response j on occasion 2 given response i on occasion 1 ($\tau^{21}_{j|i}$), multiplied by the probability of response k on occasion 3 given response j on occasion 2 ($\tau^{32}_{k|j}$).

The simple Markov chain is exceptional because ML parameter estimates can be obtained directly without using an iterative algorithm. Table 8.2

Table 8.2 *Estimation of parameters for the simple Markov model (brand purchase data)*

		$t=2$								
		$j=1$	$j=2$	Total	$\hat{\delta}^1_i$	$\hat{\tau}^{21}_{j\|i}$				
	$i=1$	581	102	683	0.69	0.85	0.15			
$t=1$	$i=2$	143	162	305	0.31	0.47	0.53			
				988						
		$t=3$								
		$k=1$	$k=2$	Total		$\hat{\tau}^{32}_{k\|j}$			$\hat{\tau}_{j\|i}$	
	$j=1$	641	83	724		0.89	0.11		$j=1$	$j=2$
$t=2$	$j=2$	115	149	264		0.44	0.56	$i=1$	0.88	0.12
								$i=2$	0.45	0.55
		$t=4$								
		$\ell=1$	$\ell=2$	Total		$\hat{\tau}^{43}_{l\|k}$				
	$k=1$	675	81	756		0.89	0.11			
$t=3$	$k=2$	111	121	232		0.48	0.52			
		$t=5$								
		$m=1$	$m=2$	Total		$\hat{\tau}^{54}_{m\|l}$				
	$\ell=1$	700	86	786		0.89	0.11			
$t=4$	$\ell=2$	86	116	202		0.43	0.57			

shows in more detail how the (conditional) probabilities are estimated. We have first presented the bivariate distribution at time points 1 and 2. The marginal distribution at $t=1$ shows that brand B ($i=2$) is purchased in 305 out of 988 cases (which is 31 per cent). The rest (683 families or 69 per cent) decided in favour of some other brand. This gives the estimates of the initial distribution at $t=1$ (The $\hat{\delta}$s). How does the purchase sequence continue if a simple Markov model applies? Given that someone did not buy brand B at $t=1$, what purchase is made at $t=2$? We see that out of the 683 families ($t=1$ and $i=1$), 581 or 85 per cent decide against brand B at $t=2$ again, whereas 102 purchasers or 15 per cent switch to brand B. The proportion of those who bought brand B at $t=1$ and stay with it at $t=2$ is 53 per cent (162 out of 305), whereas 47 per cent switch to some other brand. In Markov chain theory these conditional probabilities (the τs) are also called transition probabilities. If we proceed to compute transition probabilities from $t=2$ to $t=3$ and so on from the respective bivariate tables, we end up with the series of $\hat{\tau}$s listed in Table 8.2.

According to the simple Markov model the vector of probabilities at the second occasion, $\hat{\boldsymbol{\delta}}^2 = [0.73\ 0.27]$ in our sample, may be obtained by post-multiplying the initial probabilities, here $\hat{\boldsymbol{\delta}}^1 = [0.69\ 0.31]$, with the matrix of observed transition probabilities,

$$\hat{\mathbf{T}}^{21} = \begin{bmatrix} 0.85 & 0.15 \\ 0.47 & 0.53 \end{bmatrix}$$

The distribution at later occasions may be obtained in the same way. Therefore the simple Markov model may be represented by the path diagram of Figure 8.1. For the simple Markov chain $[\delta^1_1\ \delta^1_2] = [P_{1++}\ P_{2++}]$, $[\delta^2_1\ \delta^2_2] = [P_{+1+}\ P_{+2+}]$ etc., where $P_{i++} = \Sigma_j \Sigma_k P_{ijk}$ and $P_{+j+} = \Sigma_i \Sigma_k P_{ijk}$.

$$[\delta^1_1\ \delta^1_2] \xrightarrow{\begin{bmatrix} \tau^{21}_{1|1} & \tau^{21}_{2|1} \\ \tau^{21}_{1|2} & \tau^{21}_{2|2} \end{bmatrix}} [\delta^2_1\ \delta^2_2] \xrightarrow{\begin{bmatrix} \tau^{32}_{1|1} & \tau^{32}_{2|1} \\ \tau^{32}_{1|2} & \tau^{32}_{2|2} \end{bmatrix}} [\delta^3_1\ \delta^3_2]$$

Figure 8.1 *Path diagram of a simple Markov model: three measurements of a dichotomous variable*

The transition matrices in Table 8.2 reveal a fact that is very often observed with data of this type: they are very similar. In order to save parameters, we may therefore restrict the Markov chain to be stationary, which means that transition probabilities in formula (8.2) are homogeneous (or constant) in time:

$$\tau^{21}_{j|i} = \tau^{32}_{j|i} = \ldots = \tau_{j|i} \tag{8.3}$$

These τs, as given in the right-hand side of Table 8.2, may be estimated by the sum of all corresponding frequencies, divided by the sum of the respective marginal frequencies. In the example we have for instance: $\hat{\tau}_{1|1} = (581+641+675+700) / (683+724+756+786) = 0.88$.

The resulting stationary simple Markov chain model is a very parsimonious model in terms of parameters. Regardless of the number of panel waves, for a binary variable there are just three non-redundant parameters (one δ and 2 τs). However, is it also a good model for our example data? In order to evaluate this we compare the observed frequencies $N \times p_{ijk}$ (where N is the sample size) with the frequencies expected under the model. These are obtained by applying formula (8.2) to the parameter estimates of Table 8.2 (and again multiplying by N). A summary statistic to evaluate the fit of the model is the log of the likelihood ratio (LR) χ^2 statistic, which has convenient properties for the comparison of nested models, but which is less robust than Pearson's χ^2 statistic in the presence of scarce data. A nested model is a restricted version of a more general model. The restriction to nested models is not necessary for some information criterion statistics (Read and Cressie, 1988).

If the model were correct, the LR would be expected to be about equal to the degrees of freedom (DF). For large (random) samples a χ^2_{DF} distribution may be used to test whether the LR is significantly too large. From Table 8.3, which gives LR statistics together with degrees of freedom for various models fitted to the brand B data, we see that the simple stationary Markov model has a bad fit: the LR (= 196.53) is much larger than χ^2_{28} at the 10 per cent level (= 37.92). The poor fit of the simple stationary Markov chain, which will be found for almost every data set in the social sciences, may be due to several reasons:

Table 8.3 *Likelihood ratio (LR) statistics and degrees of freedom for various models fitted to the brand purchase data*

Model	LR	DF
Simple stationary Markov	196.53	28
Simple non-stationary Markov	187.20	22
Simple stationary Markov, second order	47.65	24
Simple non-stationary Markov, second order	42.20	16
Stationary mixed Markov, two chains	49.18	24
Non-stationary mixed Markov, two chains	23.92	12
Stationary mixed Markov, two chains, second order	16.38[1]	16
Stationary mixed Markov, three chains	28.40[1]	20
Stationary mover-stayer	71.07	26
Stationary mover-stayer with two mover chains	39.10	22
Stationary mover-stayer, second order	26.96[1]	22
Stationary independence-stayer	146.00	27
Non-stationary independence-stayer = biased coin black-and-white	123.36	24
Two-class time-homogeneous latent class	150.46	28
Two-class time-heterogeneous latent class	72.02	20
Latent stationary Markov, time-homogeneous ρs	40.13	26
Latent non-stationary Markov, time-homogeneous ρs	34.07	20
Latent stationary Markov, time-heterogeneous ρs	30.23	18
Latent Markov plus random response	20.81[1]	21
Partially latent mover-stayer	22.46[1]	24
One latent, one manifest chain	20.23[1]	22
Mixed Markov with chain turnover	23.93[1]	20
Latent mixed Markov, chain-homogeneous resp. probs.	25.29[1]	22

[1] This model fits at least at the 10 per cent level.

1 The chain is assumed to be stationary. In the present case, allowance for non-stationarity does not result in a significant improvement (the LR difference is 196.53–187.20=9.33 with 6DF).
2 The chain is assumed to be of first order. That is, knowledge of someone's state at t=2 is sufficient to predict someone's state at t=3. In a second order Markov model, transition probabilities to t=3 depend not only upon the state at t=2 but also on the state at t=1. Second-order Markov chains are not a special case of the LMMS model since transition probabilities in this model are conditional on one previous point in time only. However, they may be applied within this framework by manipulating the cross-table of $x_1, x_2, \ldots, x_T$ (Coleman, 1964; Poulsen, 1982; van de Pol and Langeheine, 1990). Since higher-order models have a rather bad image in the literature (e.g. Coleman, 1964), we will not go into details, but simply give the results of both the stationary and non-stationary version of this model in Table 8.3. As the results show, improvement in fit over the first-order model is impressive. However, the fit of these models is not satisfactory in absolute terms.

3 The model assumes the population to be homogeneous with all individuals conforming to the same process. This assumption may be unrealistic, of course. Instead, the population may be heterogeneous with two or more chains, each of which has its own dynamics. An extension presented in the next subsection develops this theme.
4 The model assumes the data to be free of measurement error. Again, this is often an unrealistic assumption in the social sciences. Below, we will therefore present the latent Markov model that allows corrections for measurement error.

To sum up, the simple Markov model has the advantage of parameter parsimony. In addition, an iterative estimation procedure is unnecessary since the parameters can be estimated directly; only a few seconds of computer time are needed to fit a model. On the other hand, the model is characterized by a series of rather restrictive and unrealistic assumptions that render it inadequate in most situations.

The mixed Markov model

To our knowledge, the first successful attempt to extend the simple Markov model to a mixture of two or more Markov chains is due to Poulsen (1982). The non-stationary mixed Markov (MM) model is given by

$$P_{ijk} = \sum_{s=1}^{S} \pi_s \, \delta^1_{i|s} \, \tau^{21}_{j|is} \, \tau^{32}_{k|js} \tag{8.4}$$

which assumes S (latent) chains, each having its own initial probabilities at t=1, $\delta^1_{i|s}$, and transition probabilities, the τs. The proportion of the population which behaves according to chain s is denoted by π_s. A stationarity restriction (formula (8.3)) may be conceived for each chain. A path diagram of a two-chain MM model is given in Figure 8.2. It is emphasized that this is a probabilistic model; the probabilities are estimated simultaneously and the model does not seek to identify which individuals are associated with each chain.

The stationary and the non-stationary two-chain mixed Markov models fit the data considerably better than the simple Markov model. The improvement in fit for the stationary two-chain MM model is about equal to the improvement obtained with the stationary second-order model. Again, however, fit is inadequate in absolute terms. To facilitate understanding of the MM model the estimated parameter values of the two-chain model are given in Table 8.4.

Under the (incorrect) assumption of a two-chain MM model the majority (77 per cent of the population) belongs to chain 1. Most of these people (82.7 per cent) decide against brand B at t = 1 and show a high repeat probability of 0.922 to stay with their behaviour at successive points in time. However, of those 17.3 per cent who initially purchased brand B, 77.8 per cent switch to some other brand on each of the successive occasions. Chain 2 depicts a somewhat different dynamic. From the members of this chain

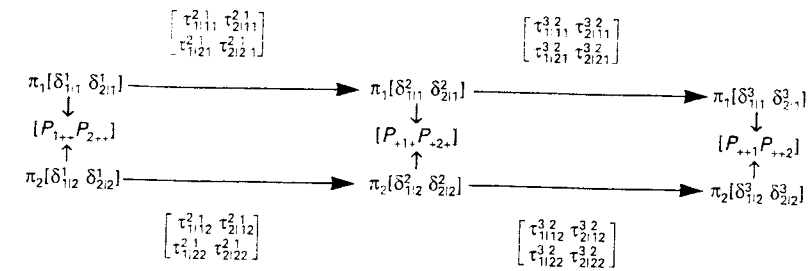

Figure 8.2 *Path diagram of a mixed Markov model: two-chain model for three measurements of a dichotomous variable*

Table 8.4 *Estimated parameter values*[1] *(and standard errors) for various mixed Markov models fitted to the brand purchase data*

Model	Chain, category	Chain proportion $\hat{\pi}$	Category proportion at t=1 $\hat{\delta}$	Transition probabilities $\hat{\tau}$ From t to t+1 Other	Brand B	From t=1 to t=5 Other	Brand B
Two-chain stationary mixed Markov	Chain 1,	0.770(0.05)					
	other		0.827(0.03)	0.922(0.01)	0.078(0.01)	0.909	0.091
	brand B		0.173(0.03)	0.778(0.06)	0.222(0.06)	0.908	0.092
	Chain 2,	0.230(0.05)					
	other		0.237(0.06)	0.475(0.11)	0.525(0.11)	0.340	0.660
	brand B		0.763(0.06)	0.270(0.04)	0.730(0.04)	0.339	0.661
Mover-stayer	Chain 1,	0.614(0.03)					
	other		0.577(0.03)	0.783(0.02)	0.217(0.02)	0.722	0.278
	brand B		0.423(0.03)	0.563(0.02)	0.437(0.02)	0.720	0.280
	Chain 2,	0.386(0.03)					
	other		0.872(0.02)	1	0	1	0
	brand B		0.128(0.02)	0	1	0	1
Independence-stayer	Chain 1,	0.534(0.02)					
	other		0.527(0.02)	0.678(0.01)	0.322(0.01)	0.678	0.322
	brand B		0.473(0.02)	0.678(0.01)	0.322(0.01)	0.678	0.322
	Chain 2,	0.466(0.02)					
	other		0.880(0.01)	1	0	1	0
	brand B		0.120(0.01)	0	1	0	1
Two-class time-homogeneous latent class	Chain 1,	0.756(0.02)					
	other		0.902(0.01)	0.902(0.01)	0.098(0.01)	0.902	0.098
	brand B		0.098(0.01)	0.902(0.01)	0.098(0.01)	0.902	0.098
	Chain 2,	0.244(0.02)					
	other		0.304(0.02)	0.304(0.02)	0.696(0.02)	0.304	0.696
	brand B		0.696(0.02)	0.304(0.02)	0.696(0.02)	0.304	0.696

[1] Parameter values of 0 and 1 are fixed by definition.

76.3 per cent bought brand B initially, and the probability to stay with brand B appears to be comparatively high (0.73). In addition, more than half of those who decided for some other brand at t=1 decide in favour of brand B at later points in time.

However, transition probabilities from t to t+1 are sometimes misleading. We have therefore also included transition probabilities from the first to the last wave. In the present case, the rows of these matrices are nearly identical within both chains. This simply tells us that the change process has reached a steady state at t=5 already. If we compare these probabilities with the initial ones we clearly see that the dynamics of the market are adverse to the newly introduced brand B within both chains (chain 1, 9 per cent versus 17 per cent; chain 2, 66 per cent versus 76 per cent).

In order to end up with an MM model with a potentially satisfactory fit we are left with several alternatives. Owing to lack of space and the availability of some more attractive models to be presented below, we will mention very briefly only two of these alternatives. Just as in the case of the simple Markov model we may specify a two-chain stationary *second-order* MM model. This model not only results in an acceptable fit (LR = 16.38, DF = 16), but also significantly improves on fit as compared with both the one-chain second-order model and the two-chain first-order MM.

Another possibility is to extend the first-order MM model from two to three chains. Again, the fit is acceptable (LR = 28.40 with 20 DF) and this model fares significantly better than the two-chain MM model. However, this model has two outliers (rather badly fitted cells).

Alternatively, the two-chain MM model may be restricted in several ways. We will comment on some special cases which are well known in the literature, although most of them do not fit well for the present data set. At the same time, this enables us to introduce some new concepts such as 'no change', 'independence' and 'random response'.

The mover-stayer model The first attempt to incorporate unobserved heterogeneity into the Markov model is due to Blumen et al. (1955). These authors proposed a mixture of two Markov chains, where movers are characterized by an ordinary stationary Markov chain, whereas stayers (the second chain, s=2) stay in their initial category with probability one across time. That is, the latter transition probabilities are restricted to be

$$\tau_{j|is} = 1 \quad \text{for } i=j \quad \text{and} \quad \tau_{j|is} = 0 \quad \text{otherwise} \tag{8.5}$$

Arranged in a matrix, these stayer or no-change probabilities constitute the identity matrix, as shown in Table 8.4. If the mover-stayer (MS) model does not fit, it may be extended with a second mover chain. Although the MS model with two mover chains does not fit adequately, there is considerable improvement upon the classical MS model as well as the two-chain MM model.

Another extension would be to make the mover chain second-order. This model fits well and is not significantly worse than the more general two-chain stationary second-order MM model (LR = 10.58 with 6 DF, whereas χ^2_6 is 10.64 at the 10 per cent level).

The independence-stayer model If the mover-stayer model fits, it may be advisable to look for more parsimonious models. One such candidate is the independence-stayer (IpS) model. It assumes that the probability of changing from time t to time $t+1$ does not depend on the category to which a person belonged at time t. All rows of the transition matrix of the movers are therefore restricted to be equal (chain 1 of IpS in Table 8.4):

$$\tau_{j|11} = \tau_{j|21} = \ldots \tau_{j|J1} \tag{8.6}$$

If the transition probabilities are also assumed to be non-stationary, we get the Converse (1974) version of the black-and-white model. This model, which has been proposed to evaluate attitude stability, postulates a dichotomy within the population: one part behaves perfectly stably over time, whereas members of the other part respond as though flipping a biased coin (maybe a different one on each occasion).

Converse (1964) originally proposed an even more restrictive version of this model, where people were assumed to flip an unbiased coin at every occasion, i.e. to respond randomly. This model would require the following restrictions for the random response chain: $\delta^1_{j|1} = 1/J$; $\tau_{j|i1} = 1/J$.

The classical latent class model Consider the S-chain non-stationary MM model discussed earlier. This model says that the probability of being in category j at some time $t+1$ depends not only on the category i at the previous occasion but also on chain membership. If we drop the local dependence assumption within every transition matrix (that is, formula (8.6) holds) the result is a model with S independence chains which is Lazarsfeld's (1950) classical latent class (LC) model:

$$P_{ijk} = \sum_{s=1}^{S} \pi_s \, \delta^1_{i|s} \, \delta^2_{j|s} \, \delta^3_{k|s} \tag{8.7}$$

given here for three measurements of one variable. The classical latent class model may be interpreted as a heterogeneity model: the S latent classes in the population are assumed to have different distributions, $\delta^1_{i|1}, \ldots, \delta^1_{i|S}$ for occasion (indicator) 1, $\delta^2_{j|1}, \ldots, \delta^2_{j|S}$ for occasion 2, and so on. The observed distribution is a mixture, or average, of these distributions with weights π_s. Equation (8.7) is the time-heterogeneous version of this model. Of course, time homogeneity may be assumed by imposing restrictions $\delta^1_{i|s} = \delta^2_{i|s} = \delta^3_{i|s}$. For illustrative purposes, parameter estimates of the latter version are included in Table 8.4 in terms of a restricted MM model, the two-class time-homogeneous LC model.

To sum up, the mixed Markov model turns out to be rather general. Not only does it explicitly take into account unobserved population heterogeneity, but it also contains a variety of models well known in the literature as special cases, such as the simple Markov model (which is MM with $S=1$), the mover-stayer model, the black-and-white models and the classical latent class model. In addition, many other models (such as mover-independence, mover random response etc.) may be devised (van de Pol and Langeheine, 1990). Just as LC models are sometimes called local independence models, so MM models may be conceptualized as local dependence LC models. These models are latent because an unobserved variable with S categories is introduced in addition to the manifest xs.

A restrictive aspect of MM models, however, is that membership in chain s is permanent across time, i.e. there is no turnover between chains. Another drawback is that (with the exception of LC versions) MM models may not be given an error in measurement interpretation, which is a special feature of the latent Markov model that will be presented in the next section.

Finally, all of the MM models mentioned so far may be estimated using the PANMARK program. Since estimation is done iteratively, a solution with a local (though not global) maximum of the log-likelihood may be found. Poulsen (1982) reports such a solution (LR = 69.03 instead of our 49.18) for the two-chain stationary MM model. We therefore urge users not to rely on results obtained from a single set of starting values (PANMARK allows for default or user-specified starting values). In addition, identifiability of a specific model should be checked carefully. Again, PANMARK provides several ways in which this may be done. Both of these issues are addressed in detail by van de Pol and Langeheine (1989).

The latent Markov model

The latent Markov (LM) model considered in this section is due to Wiggins (1955; 1973):

$$P_{ijk} = \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \delta^1_a \, \rho^1_{i|a} \, \tau^{21}_{b|a} \, \rho^2_{j|b} \, \tau^{32}_{c|b} \, \rho^3_{k|c} \tag{8.8}$$

This model assumes that the whole population changes according to one Markov chain, but measurement error causes the observed cross-table to diverge from what would be observed if reliability was perfect. The model thus allocates a subject into one of A latent classes at $t=1$ which are characterized by response probabilities ($\rho^1_{i|a}$). For panel data the number of latent classes, A, is usually assumed to be equal to the number of manifest response categories, J. Moreover, latent classes are ordered so that we get high values for the diagonal elements $\rho^1_{i|a}$ (with $a=i$) in the $A \times J$ matrix of response probabilities $\mathbf{R}$. Change from one point in time to the next is captured by latent transitions, the τs. That is, someone may stay in a class ($b=a$) or switch to some other class ($b \neq a$). A path diagram is given in Figure 8.3.

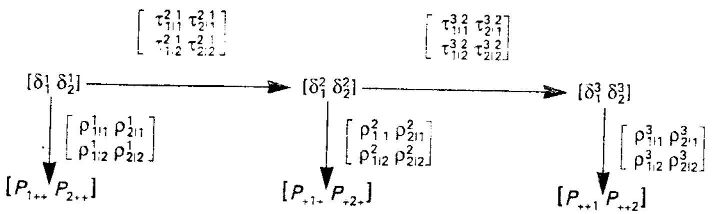

Figure 8.3 *Path diagram of a latent Markov model: three measurements of a dichotomous variable*

As with MM models, time homogeneity may be assumed for transition probabilities τ, giving a latent stationary Markov chain. Moreover, in the LM model, time homogeneity may also be assumed for response probabilities. In general, time heterogeneity of the τs does not cause any problems, but, for three waves only, the ρs have to be time homogeneous in order to render a model identifiable.

For the brand B data the latent variable may be labelled 'brand loyalty', buying behaviour being a fallible indicator. None of various two-class LM models fits (see Table 8.3). The LM model with both time-heterogeneous ρs and τs is not identified. Just as with MM models, relaxing time homogeneity in one way or another does not significantly improve the fit.

Nevertheless, we have included estimated parameter values of the stationary LM model in Table 8.5 for illustrative purposes. Comparison with the results of the simple Markov model reveals some interesting features of the LM model. First, extension by two parameters only (the ρs) results in a dramatic improvement in fit (from 196.53 to 40.13: see Table 8.3). Second, response probabilities indicate that the implicit assumption of perfect reliability in the simple Markov model (model M) is incorrect; the class 1 $\rightarrow$ other and class 2 $\rightarrow$ brand B probabilities are each significantly less than unity. Third, this results in overestimation of change by model M; the off-diagonal transitional probabilities for model M (see Table 8.2) are larger than the off-diagonal transitional probabilities for the LM model.

If the LM model fits, the conclusion may be that some real change takes place. However, before accepting this hypothesis, the LM model should be confronted with the rival hypothesis of no latent change. This hypothesis requires the transition matrix to be the identity matrix (or a latent stayer matrix). In this case, we have one latent variable with A categories only, and the τs are dropped from formula (8.8), which reduces to

$$P_{ijk} = \sum_{a=1}^{A} \delta_a \, \rho^1_{i|a} \, \rho^2_{j|a} \, \rho^3_{k|a} \qquad (8.9)$$

which, again, is the same as the classical latent class model in equation (8.7). The present notation, however, suggests a measurement error interpretation

Table 8.5 *Estimated parameter values*[1] *(and standard errors*[2]*) for two latent Markov models*[3] *fitted to the brand purchase data*

Model	Class	Class proportions $\hat{\delta}$	Response probabilities $\hat{\rho}$		Latent transition probabilities $\hat{\tau}$ From t to $t+1$			From $t=1$ to $t=5$		
			Other	Brand B	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3
Latent Markov	1	0.667(0.02)	0.936(0.01)	0.064(0.01)	0.982(0.01)	0.018(0.01)	-	0.949	0.051	-
	2	0.333(0.02)	0.213(0.03)	0.787(0.03)	0.187(0.02)	0.813(0.02)	-	0.548	0.452	-
Latent Markov with random response class	1	0.535(0.05)	0.966(0.02)	0.034(0.02)	0.968(0.04)	0.002(0.02)	0.030(0.06)	0.915	0.008	0.077
	2	0.138(0.05)	0.100(0.08)	0.900(0.08)	0.089(0.16)	0.900(0.14)	0.011(0.28)	0.305	0.656	0.039
	3	0.327(0.09)	0.5	0.5	0.237(0.09)	0.004(0.10)	0.759(0.17)	0.628	0.011	0.361

[1] Parameter values of 0.5 are fixed by definition.
[2] Standard errors are obtained from the information matrix.
[3] Both models assume time-homogeneous ρs and τs.

of the classical LC model, with only one latent distribution, δ_a, which is unreliably measured at three occasions (or with three indicators). Latent Markov models may therefore also be called latent class models with latent change.

Since the LM model with $A=J$ classes did not fit the brand B data, we will consider two extensions. Poulsen (1982) reports a satisfactory fit for the three-class unrestricted stationary LM model (LR = 20.58, DF = 20). Re-estimation with the PANMARK program revealed that this model is not identified. Poulsen's results suggest, however, that we estimate a model with the third class consisting of people who respond randomly, or who buy both brand B and other brands simultaneously. That is, we fix the response probabilities of this class at 0.5. The fit of this model is excellent, but transition probabilities are estimated with large standard errors (Table 8.5) (latent Markov with random response: LMRr).

Probably the most surprising result is that about one-third of the population is classified as giving random response (or buying both brand B and other brands) at $t=1$. Measurement is almost perfect for class 1 which contains buyers of all other brands. As compared with the LM model, brand B purchasers (class 2) are also measured more reliably now. Both sets of transition probabilities (from one point in time to the next and from the first to the last wave) reveal that there is considerable change over time in favour of switching to some other brand. Nearly one-third of the brand B buyers at $t=1$ do not show brand loyalty at $t=5$. Nearly two-thirds of those initially classified random respondents switch to some other brand at $t=5$. The latent distribution at $t=5$ (the vector of estimates is $\hat{\boldsymbol{\delta}}^5 = [0.736\ 0.099\ 0.165]$) shows total net latent change to be 20.1 per cent in favour of some other brand. Note that this is mostly due to those who gave random response (or bought both brand B and other brands) at the first wave.

To sum up, the main feature of the latent Markov model, which postulates one latent chain to hold for the entire population, is to correct manifest responses for measurement error. Consequently, transition probabilities are latent and allow for turnover from one latent class to another across time. In general, comparison with the simple (manifest) Markov model shows that the latter overestimates both reliability and change. We will return to this issue below. Again, the LM model contains the classical latent class model as a special case which is LM with the transition matrix restricted to be the identity matrix.

As long as the number of classes is specified to be equal to the number of manifest response categories, we have never encountered any identifiability problems, not even with a non-stationary latent Markov chain. However, these may occur with time-heterogeneous response probabilities or if the number of classes is allowed to be greater than the number of response categories.

Of course, latent Markov chains can also be measured via multiple indicators. Consider for a moment a latent Markov model for one indicator and six occasions with non-stationary transition probabilities. A path diagram of this model is given in Figure 8.4a with matrices of transition probabili-

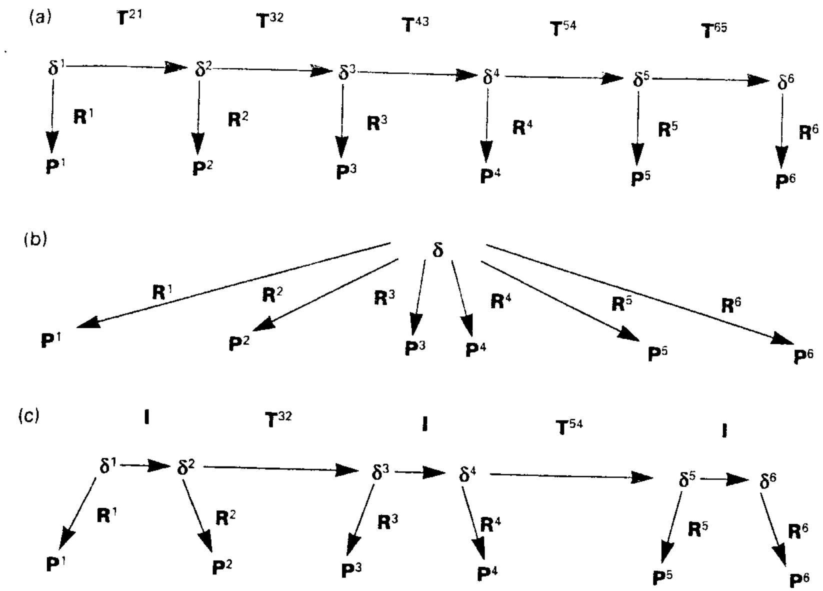

Figure 8.4. *Path diagrams for (a) a latent Markov model (one indicator, six occasions) (b) a classical latent class model (c) a multiple-indicator latent Markov model (two indicators, three occasions)*

ties from time $t-1$ to time t designated by $\mathbf{T}^{t,t-1}$, matrices of response probabilities at time t designated by $\mathbf{R}^t$, and with $\mathbf{P}^t$ designating the observable distribution vector at time t. If all $\mathbf{T}^{t,t-1}$ are restricted to be the identity matrix $\mathbf{I}$ (i.e. with diagonal elements equal to 1 and other elements 0), the latent Markov model reduces to a classical latent class model (formula (8.9)) with one latent variable that does not change over time. The respective path diagram in Figure 8.4b shows that variables at all time points are considered as (fallible) indicators of one latent variable. Now think of a situation where we have two indicators of one latent variable for three occasions. This can be modelled as a latent non-stationary Markov chain by using one latent variable for each indicator and restricting some of the matrices of transition probabilities between these latent variables to be the identity matrix, as shown in Figure 8.4c. Parameter estimates will be less sensitive to departures from the Markov assumption if multiple indicators are used.

The latent mixed Markov model

Model enthusiasts might be interested in whether there is an even more general model that contains both the mixed Markov and the latent Markov model as special cases. In fact, there are several ways to derive such a model. The version we prefer, the latent mixed Markov (LMM) model in Langeheine and van de Pol (1990), is given by

$$P_{ijk} = \sum_{s=1}^{S} \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \pi_s \, \delta^1_{a|s} \, \rho^1_{i|as} \, \tau^{21}_{b|as} \, \rho^2_{j|bs} \, \tau^{32}_{c|bs} \, \rho^3_{k|cs} \tag{8.10}$$

This extends the single-chain LM model to a mixture of S latent chains, having one additional set of parameters, the πs, which are the proportions of the S chains. Within each chain, the chain variable (with index s) is added as a conditioning variable to the rest of the parameters. A path diagram of the LMM model would look like Figure 8.2, with matrices of response probabilities added on the arrows from δs to Ps, as in Figure 8.3.

As Table 8.6 shows, any S-chain A-class LMM model may be formulated as a one-chain LM model with $A' = S \times A$ classes with certain transition probabilities defined to be structural zeros.

All models considered above are special cases of the LMM model. This is quite obvious for the LM model, which is LMM with S=1. The MM model is obtained by specifying perfect measurement within all classes (e.g. $\rho_{i|a} = 1$ if a=i and $\rho_{i|a} = 0$ otherwise).

In principle, the LMM model opens a wide avenue for specifying new models unknown in the literature so far. However, this generality comes at a price. That is, an LMM model may not be identified unless some restrictions are imposed on the general version. First experience from a few data sets has shown that measurements from at least six waves are necessary for the general version to be identified. Identification problems can be overcome by using more than one indicator for each occasion.

For the five-wave brand B data several two-chain two-class models have been considered. The general version is not identified in this case, nor is a model that restricts the transition matrix of one of these chains to be the identity matrix. Note that this model may be called a latent mover-stayer model owing to incorporation of measurement error into both chains, as opposed to the classical mover-stayer model. We therefore decided to specify perfect measurement for the stayers, which reduces the latent mover-stayer model to a partially latent model with latent movers but manifest stayers. This model fits extremely well and significantly better than the one-chain

Table 8.6 *Latent transition parameters*[1] *in a two-chain two-class latent Markov model in terms of* $A' = 2 \times 2$ *classes*

				Time $t+1$				
			$b'(t+1)$	1	2	3	4	Total
			Chain s	1	1	2	2	
	$a'(t)$	Chain s	Class a / Class b	1	2	1	2	
	1	1	1	$\tau_{1\|11}$	$\tau_{2\|11}$	0	0	1
	2	1	2	$\tau_{1\|21}$	$\tau_{2\|21}$	0	0	1
Time t	3	2	1	0	0	$\tau_{1\|12}$	$\tau_{2\|12}$	1
	4	2	2	0	0	$\tau_{1\|22}$	$\tau_{2\|22}$	1

[1] Structural zeros are fixed by definition.

Source: adapted from Langeheine and van de Pol, 1990

Table 8.7 *Estimated parameter values*[1] *(and standard errors) for the partially latent mover-stayer model (brand purchase data)*

Chain	Class	Chain proportions $\hat{\pi}$	Class proportions $\hat{\delta}$	Response probabilities $\hat{\rho}$: Other	Brand B	Latent transition probabilities $\hat{\tau}$: From t to t+1: Class 1	Class 2	From t=1 to t=5: Class 1	Class 2
1		0.824(0.11)							
	1		0.587(0.05)	0.932(0.03)	0.068(0.03)	0.972(0.01)	0.028(0.01)	0.924	0.076
	2		0.413(0.05)	0.308(0.05)	0.692(0.05)	0.233(0.03)	0.767(0.03)	0.626	0.374
2		0.176(0.11)							
	1		0.778(0.15)	1	0	1	0	1	0
	2		0.222(0.15)	0	1	0	1	0	1

[1] Parameter values of 1 and 0 are fixed by definition.

two-class LM model which is extended by the two additional parameters for perfectly stable stayers (see Tables 8.5 and 8.7). The model also fits significantly better than the classical mover-stayer model to which it adds two parameters to capture measurement error in the mover chain.

The estimated parameter values for the partially latent mover-stayer model are given in Table 8.7. According to this model, perfectly stable stayers of all other brands ($17.6 \times 0.778 = 13.7$ per cent) and perfectly stable brand B purchasers ($17.6 \times 0.222 = 3.9$ per cent) add up to 17.6 per cent. All blame for measurement error now goes to chain 1. This is the reason for the somewhat lower reliabilities as compared with the respective values for the LM and LMRr models (see Table 8.5). Again, both sets of transition probabilities show change over time towards being adverse to brand B. The latent distribution at the last wave is $\boldsymbol{\delta}^5$: [0.659 0.164 0.137 0.039]. This model thus says that there is a total net loss of 0.659–0.484 = 17.5 per cent of brand B market share during the observation period. There is no steady state at t=5. The model thus predicts that loss of brand B market share will continue.

Before concluding this section we shall briefly mention some additional models, all of which are LMM with some restrictions added. One such candidate is a mixture of one latent and one manifest chain. This model simply removes the no-change restriction of the partially latent mover-stayer model. But the two parameters added to the latter model do not significantly increase the fit for the brand purchase data.

In commenting on the mixed Markov model we mentioned the restrictive aspect of constant chain membership across time. Now that we can formulate an MM model as an LMM model, it is easy to incorporate turnover between chains into the MM model by removing (some of the) zero restrictions for transition probabilities in Table 8.6 (while fixing response probabilities at zero and one). For the two-chain MM model with chain turnover listed in Table 8.3 we removed four zero restrictions on the transitions, i.e. the transition matrix looks like

$$\begin{bmatrix} \tau & \tau & 0 & \tau \\ \tau & \tau & \tau & 0 \\ 0 & \tau & \tau & \tau \\ \tau & 0 & \tau & \tau \end{bmatrix}$$

Moreover, in this four-class latent Markov model the response probability matrices are restricted to

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This model also fits quite well and significantly improves on the fit of the traditional MM model, to which it adds four parameters.

Another possibility is to assume the response probabilities of one chain to be equal to the corresponding response probabilities of the other chain, without imposing such equality restrictions on transition probabilities. We found this model to fit well for the brand purchase data. Likewise, we may specify transitions to be chain homogeneous without imposing such restrictions on response probabilities. The latter model is not identified for the brand purchase data, however.

In conclusion, it should have become evident that, though simple in principle, the LMM model allows for complex extensions of both MM and LM models. However, some models that may be conceptually valid may not be identifiable for one indicator per occasion, especially if the number of occasions is small.

Stability and change

Now that we have considered quite a variety of models for the brand purchase data, we will comment briefly on the different types of stability and change implied by these models (for details, see Langeheine and van de Pol, 1990). Table 8.8 gives estimated proportions for some selected models.

First consider models from the simple Markov to the latent class formulation (M to LC) listed across the top of the table. In these models, all change is considered to be manifest (and likewise stability). In comparison with the respective proportions in the data (where the two cells with response patterns 11111 and 22222 indicate stability and the rest correspond to change), models M, MM and LC overestimate manifest change. In mover-stayer models this proportion equals that in the data, because no-change data cells are fitted perfectly. In MS models we distinguish perfect stability for stayers from stability in mover chains. Note that any Markov chain except for a stayer

Table 8.8 *Estimated proportions of stability and change in some selected models (brand purchase data)*

				Model				
	Data	M	MS	MM	LC	LC′	LM	pLMS
Perfect stability			0.386					0.176
Stability	0.528	0.444	0.142	0.513	0.492	0.492	0.766	0.550
true stability						0.491	0.490	0.322
measurement error						0.001	0.276	0.228
Change	0.472	0.556	0.472	0.487	0.508		0.234	0.274
true change							0.116	0.105
measurement error						0.508	0.118	0.169
Total error						0.509	0.394	0.397

M, simple Markov; MS, mover-stayer; MM, stationary two-chain mixed Markov;
LC, two-class time-homogeneous latent class; LC′, error of measurements interpretation of LC;
LM, two-class latent stationary Markov with time-homogeneous response probabilities;
pLMS, partially latent mover-stayer

or random response chain contains a certain amount of stability despite the fact that it may be called a mover chain.

In models that allow for an 'error of measurements' interpretation (the models LC' to pLMS listed across the top of Table 8.8), both stability and change may be broken down into a true and an error proportion. In a latent stayer model (LC') there is no change by definition. Consequently, all change is considered as error in this model. Figures for the latent Markov model as well as the partially latent mover-stayer model both demonstrate the well-known fact that stability is underestimated while change is overestimated if the problem of measurement error is neglected. In both of these models, the proportion of total error sums to nearly 40 per cent with approximately 50 per cent true stability and about 11 per cent true change.

Note that all quantities in Table 8.8 give total proportions. Net change, which refers to change in marginal distributions, should therefore be distinguished from total change.

Models for several subpopulations

Very often multi-way tables of consecutive measurements of one target variable are available not only for a single sample but also for subsamples that are defined by one or more qualitative variables such as gender, age, or education. The additional variables may have been recorded because the subgroups so defined are expected to differ in their dynamics on the target variable. In such cases it is advisable to perform a simultaneous analysis for all subgroups instead of separate analyses for each subgroup or even a single analysis of the total sample pooled across subgroups.

To enable this sort of analysis, the latent mixed Markov model (8.10) is extended by a parameter γ_h that refers to the proportion of each of the H subpopulations, and all other parameters are considered conditional on subpopulation h. This brings us back to the general latent mixed Markov model for several subpopulations which was outlined at the outset of this chapter (equation (8.1)); we have reached the top of the model hierarchy.

This model opens a Pandora's box. All the models considered above may be fitted into this framework. At one extreme, complete heterogeneity of subsamples may be permitted, leaving each subsample with its own parameters. This would be equivalent to fitting a model to each subsample separately. At the other extreme, it is possible to assume homogeneity across subsamples. Between these two extremes many across-group restrictions may be devised that result in partially homogeneous models. In addition, within-group restrictions may be specified with, for example, a mover-stayer model for one subgroup but an independence-stayer model for another. If there are two or more variables defining subgroups, these variables may be combined into one composite variable (for example, two binary variables define H=4 subgroups).

However, sample sizes for subpopulations will be smaller, resulting in less accurate parameter estimates (larger standard errors). Put differently, a

model with complete across-group heterogeneity requires H times the number of parameters of the corresponding model for the pooled sample (excluding γ_h). More realistic models may therefore require some across group equality restrictions. If theory is lacking, this requirement may be difficult to achieve, especially in view of the multitude of potential models. In van de Pol and Langeheine (1990) we have therefore proposed a procedure for the selection of appropriate equality constraints across subgroups.

A panel analysis on two subpopulations

In this section we will consider a sample of people who reported some work disability in 1971, and who were reinterviewed about self-assessed work disability in 1972 and 1974 (Table 8.9). Bye and Schechter (1986), who proposed another method for estimating the parameters of the latent Markov model, report results for the pooled sample and for males and females considered as separate groups.

Some results[1] are given in Tables 8.10 and 8.11. The models in Table 8.11

Table 8.9 *Contingency tables for males and females having some work disability[1] in 1971*

Time 1971	1972	1974	Sample Males	Females	All cases
1	1	1	79	140	219
1	1	2	7	34	41
1	2	1	8	26	34
1	2	2	13	26	39
2	1	1	14	31	45
2	1	2	1	17	18
2	2	1	23	24	47
2	2	2	69	76	145

[1] Categories: 1, severely disabled; 2, not severely disabled or no longer disabled.

Source: Bye and Schechter, 1986

Table 8.10 *Likelihood ratio statistics (LR) and degrees of freedom (DF) for latent Markov models fitted to the work disability data*

Subgroups	Stationarity	Across-group equality constraints: Class prob. $\hat{\delta}$	Response prob. $\hat{\rho}$	Transition prob. $\hat{\tau}$	LR	DF
All cases	Yes	–	–	–	3.05	2
Male/female	Yes	No	No	No	11.54	4
Male/female	Fem. only	No	No	No	0.07	2
Male/female	Fem. only	Yes	No	No	3.01	3
Male/female	Fem. only	No	Yes	No	7.68	4
Male/female	Yes	No	No	Yes	12.37	6
Male/female	Yes	Yes	Yes	Yes	34.16	9

Table 8.11 *Estimated parameter values (and standard errors) for the latent stationary Markov model on work disability*

Subpop., class	Subpop. proportions $\hat{\gamma}$	Class proportions $\hat{\delta}$	Response probabilities $\hat{\rho}$		Latent transition probabilities $\hat{\tau}$			
					From t=1 to t=2		From t=1 to t=5	
			Severe	Other	Class 1	Class 2	Class 1	Class 2
All cases								
Class 1		0.574(0.03)	0.914(0.02)	0.086(0.02)	0.911(0.02)	0.089(0.02)	0.843	0.157
Class 2		0.426(0.03)	0.083(0.04)	0.917(0.04)	0.148(0.04)	0.852(0.04)	0.261	0.739
Males	0.364(0.02)							
Class 1		0.501(0.05)	0.967(0.02)	0.033(0.02)	0.902(0.04)	0.098(0.04)	0.831	0.169
Class 2		0.499(0.05)	0.017(0.05)	0.983(0.05)	0.177(0.05)	0.823(0.05)	0.306	0.694
Females	0.636(0.02)							
Class 1		0.637(0.05)	0.875(0.03)	0.125(0.03)	0.925(0.04)	0.075(0.04)	0.863	0.137
Class 2		0.363(0.05)	0.127(0.05)	0.873(0.05)	0.105(0.07)	0.895(0.07)	0.192	0.808

are latent stationary Markov with time-homogeneous response probabilities. If males and females are pooled, the model fits well (Table 8.10: LR = 3.05 with 2 DF). The same parameter estimates are found if males and females are not pooled, but all parameters are set equal for men and women. However, for these non-pooled data the fit of the LM model appears to be very bad (LR = 34.16 with 9 DF). Even if all parameters are assumed different for men and women we find an unsatisfactory fit (LR = 11.54 with 4 DF). Inspection of the standardized residuals for this model shows that the poor fit is due largely to the males subgroup. When the stationarity assumption is relaxed, males' disability status turns out to have changed much more in the first period than in the second, and also more than females' disability status has changed in any period. Therefore a model with time-heterogeneous transition probabilities for males only is chosen, which fits well (LR = 0.07, DF = 2). The stationarity assumption for females is not relaxed because the possible bias caused by this assumption may be outweighed by lower standard errors of the estimates.

There is a clear message from these analyses: pooling of subgroups may obscure heterogeneity which can be detected in a simultaneous analysis of several subpopulations.

Our results are consistent with Bye and Schechter's conclusion that response error is less for males than for females. However, our methodology allows a single likelihood ratio test of this hypothesis by constraining response probabilities to be equal across groups (LR = 7.68–0.07 = 7.61 with 2 DF). In interpreting their results, Bye and Schechter (1986:378) note:

> The difference . . . for males and females might be explained by the work-related nature of the disability questions in the surveys. Given that women are more likely to be marginally attached to the labour force than men, women might have greater difficulty in deciding whether they are unable to work altogether.

Concluding comments

The aim of this chapter is to demonstrate the main features of discrete-space, discrete-time Markov modelling. In so doing, we have decided against the more traditional approach of starting with the theoretical, mathematical and statistical background and ending with the application to a substantive example. Instead, we have followed a bottom-up strategy that starts with the most simple model. The weaknesses of this model may be easily shown by extending it to a mixed Markov model or a latent Markov model, both of which turn out to be special cases of the more general latent mixed Markov model. Finally, we have shown how the latter model may be extended to cope with the situation where several subpopulations instead of a single one are the focus of the analysis. To emphasize that all of these models are latent class models, be it with or without the assumption of local independence, we refer to them as mixed Markov latent class (MMLC) models.

In general, we have tried to keep the reference list short by giving some key references only. This is not to say that Markov models have found little interest so far. Extensive references, both with respect to theory and applications, may be found in Bartholomew (1981), van de Pol and Langeheine (1989; 1990) and Langeheine and van de Pol (1990).

Although the methodology presented in this chapter can be very effective it is not, of course, without problems. Some of these problems may be overcome, whereas others are yet to be solved. One serious problem that belongs within the first category is model identifiability. For those familiar with log-linear modelling, it may be surprising that a Markov model may turn out to be non-identifiable, even though there are surplus degrees of freedom. Those who are familiar with classical latent class analysis are aware of this problem, however. In classical LC models a latent categorical variable is assumed to explain all associations among the manifest variables. Put differently, all manifest associations vanish, given class membership. For this reason, these models are called local independence models. Markov models differ from classical LC models either by allowing local dependence (association of manifest variables in mixed Markov models) or by restricting local independence to indicators of the same latent variable (latent Markov models). Hence, it should not be surprising that identifiability problems may arise. In general, these may be detected by inspection of the diagnostics provided by the PANMARK program.

In building up the hierarchy of models presented in this chapter we have fitted a number of different models to the brand choice data. On the one hand, this allowed us to demonstrate the strengths and weaknesses of these models. On the other hand, such an exploratory procedure may be denounced as data dredging or as bad science. Those who oppose this approach would argue in favour of using substantive theory as a guide in selecting (potentially rival) models. We do not reject this position. However, it carries the risk of missing a well-fitting model. Each of the many models is a legitimate way to look at the data. One or the other may be better suited to a specific research problem. We would, however, emphasize the important problem of spurious 'change' due to measurement error.

If attention is confined to a hierarchy of nested models, there is a convenient criterion for identifying the preferred model: the log of the likelihood ratio (LR). If non-nested models are to be compared, a more general criterion is needed, such as the Akaike information criterion (AIC) or the Bayesian information criterion (BIC) (see Read and Cressie, 1988). Both of these are a function of the log-likelihood, the number of parameters and (BIC only) the sample size. (In fact, there are many recent publications from different areas of application where authors rely fully on one of these criteria instead of formal goodness-of-fit tests.) For illustrative purposes we have given BIC together with some additional information in Table 8.12 for those models that fit the brand choice data well. If a choice has to be made, then BIC points to the partially latent mover-stayer model.

A problem still to be tackled is the unrealistic assumption that a panel is

Table 8.12 *LR statistic, DF, log-likelihood, number of parameters, and Bayesian information criterion for selected models from Table 8.3*

Model	LR	DF	Log-likelihood	Number of parameters	BIC
pLMS	22.46	24	−2292.83	7	4633.93[1]
One latent, one manifest chain	20.23	22	−2291.71	9	4645.48
LMM, chain-homogeneous resp. prob.	25.29	22	−2294.24	9	4650.54
MS, second order	26.96	22	−2295.08	9	4652.22
LMRr	20.81	21	−2292.00	10	4652.96
MM with chain turnover	23.93	20	−2293.56	11	4662.97
MM, three chains	28.40	20	−2295.79	11	4667.43
MM, two chains, second order	16.38	16	−2289.79	15	4683.02
The data	0.0	0	−2281.60	31	4776.97

[1] Best model according to BIC, which is given by − 2 (log-likelihood) + q log(N), where q is the number of parameters.

a closed system which no one leaves or joins. Typically, there will be some non-response or attrition at each panel wave. This problem is often circumvented by including in the analysis only subjects with a complete record at the expense of (possibly selective) loss of information from those with incomplete observations. To further complicate the issue, in some panel designs new members join the panel at later points in time.

Little and Rubin (1987: section 11.6) discuss various models for two occasions, including a third variable: (non-)response at the second occasion. Some of these cannot be estimated. In other cases the response variable can be ignored, because the missing cases are 'missing completely at random'. From those models that remain, the 'missing at random' assumption (randomness conditional on some variable) is an interesting option.

For Markov models considered in this chapter, missing values may be handled by introducing 'missing' as another category. For MMLC models with response probabilities, a more parsimonious model will be obtained if no latent class is assumed for the missing cases. This means that the observed scores are assumed 'missing at random' given the relevant latent variable. A test for 'missing completely at random' may be performed by assuming the probability of 'missing' equal for all classes of the relevant latent variable. For MM models such a test involves conditioning on the previous occasion of a variable. Where there are complicated patterns of inflow and outflow of panel members at successive occasions, an LM model for several subsamples (complete cases and several categories of incomplete cases) might be useful to test whether transition probabilities of complete cases may be considered equal to those of incomplete cases.

Finally, some comments about software are in order. As we have mentioned earlier, all models considered in this chapter were fitted by using the PANMARK program. Although PANMARK has been designed especially to fit Markov models, it can be used to fit a large number of classical latent class models that have become quite popular in the social sciences in the 1980s (see Langeheine. 1988 for a recent review). The trick is simply to replace time points, which are the 'variables' in a T-wave contingency table, by items or variables of a T-way contingency table. PANMARK may then be used to estimate all LC models that can be handled by the probably more widely known MLLSA program of Clogg (1977), and many more including, for example, mixtures of locally (in)dependent classes. In addition, PANMARK provides some information which is lacking in MLLSA but which may be very helpful in assessing the adequacy of a specific model (e.g. standard errors, first-order derivatives, detailed identification test).

Notes

The views expressed in this chapter are those of the authors and do not necessarily reflect the policies of the Netherlands Central Bureau of Statistics.

1 Note that our results differ somewhat from those reported by Bye and Schechter. For the model of all cases, for instance, we find better-fitting parameter estimates than Bye and Schechter did (LR = 3.05 versus LR = 4.75). The same holds for the 'no-no-no' model without equality restrictions between men and women, which was also fitted by Bye and Schechter. This may be due to more iterations performed in our computations or to a better performance of the EM algorithm used in the PANMARK program.

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